PROGRAMMING LOGIC AND DESIGN

(CpE 02)

1. Write a function createArithmeticSeq that prompts the user to input two numbers, first and diff. The function then creates a one-dimensional array of 16 elements ordered in an arithmetic sequence. It also outputs the arithmetic sequence. For example, if first = 21 and diff = 5, the arithmetic sequence is 21 26 31 36 41 46 51 56 61 66 71 76 81 86 91 96.

2. A program that will input 5 numbers and determine the highest and lowest number. The computer will then display the entered numbers in ascending and descending order.

3. Create a program that will accept a decimal value input and then choose to either convert it to one of the following number systems:

   1 – Decimal to Binary
   2 – Decimal to Octal
   3 – Decimal to Hexadecimal

   USE ARRAYS TO FORM THE NEWLY CONVERTED NUMBER

   Sample Output:
   Enter a positive decimal value: 14
   value to which number system:
   1 – Decimal to Binary
   2 – Decimal to Octal
   3 – Decimal to Hexadecimal
   Enter choice: 1
   Equivalent of 14 in binary is 1110

4. Write a program that determines if the triangle is equilateral, isosceles or scalene. The values for the 3 sides will be from the user.

5. Write a Java program to check if a number is a palindrome in Java? (121 is a palindrome, 321 is not).

6. Write a program that determines the day number 1 to 366 in a year for a date that is provided as input data. As an example, January 1, 1994 is day 1. December 31, 1993 is day 365. December 31, 1996 is day 366, since 1996 is a leap year. A year is a leap year if it is divisible by four, except that any year divisible by 100 is a leap year only if it is divisible by 400. Your program should accept the month, day and year as integers.

7. Write a program in Java to find out if a number is prime or composite?

8. Write a Java program that outputs the Greatest Common Divisor of 2 integers.

9. Write a program to process a collection of daily high temperature. Your program should count and print the number of hot days(high temperature 85 or higher), the number of pleasant days(high temperature 60-84), and the number of cold days(high temperatures less than 60. It should also display the category of each temperature. Test your program on the following data:
55 62 68 74 59 45 41 58 60 67 65 78 82 88 91 92 90 93 87 78 79 72 68 61 59

10. Write a program to read a list of exam scores (integer percent scores, in the rage 0 to 100) and output the total number of grades and then number of grades in each lettergrade category (90 to 100 =A, 80 to 89=B, 70 to 79 = C, 60 to 69 = D, and 0 to 59 = F). The end of the input is indicated by entering a negative score as a sentinel value (the negative value is used only to end the loop, so do not use it in the calculations).

For example, if the input is : 98 87 86 85 85 78 73 72 72 72 70 66 63 50 -1 the output would be Total number of grades = 14 Number of A's = 1 Number of B's = 4 Number of C's = 6 Number of D's = 2 Number of F's = 1